Section 6

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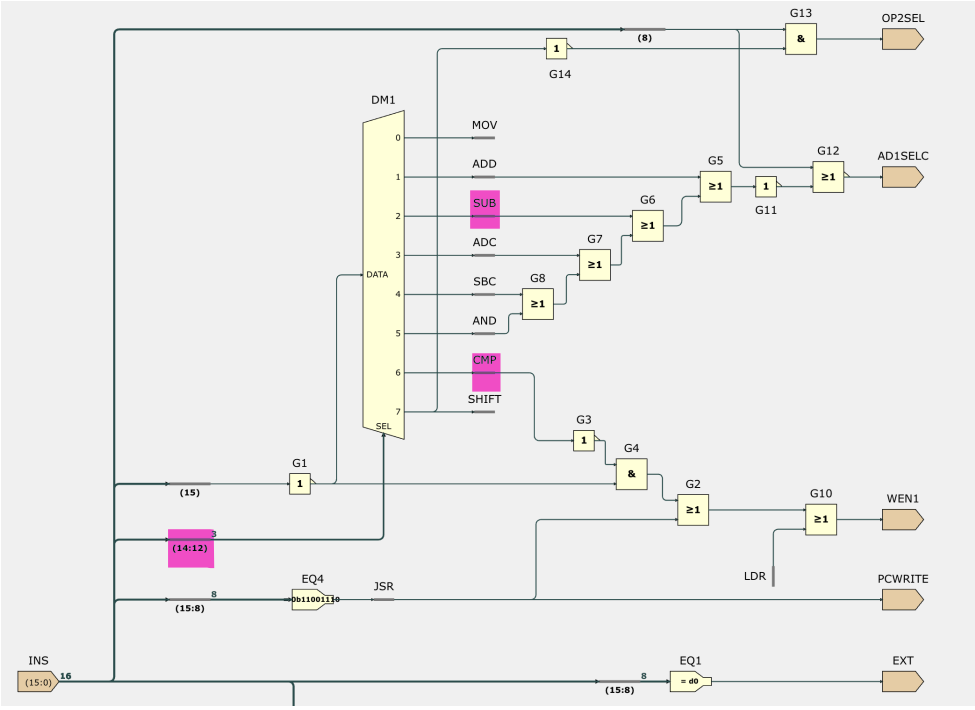
6 CMP and AND instructions

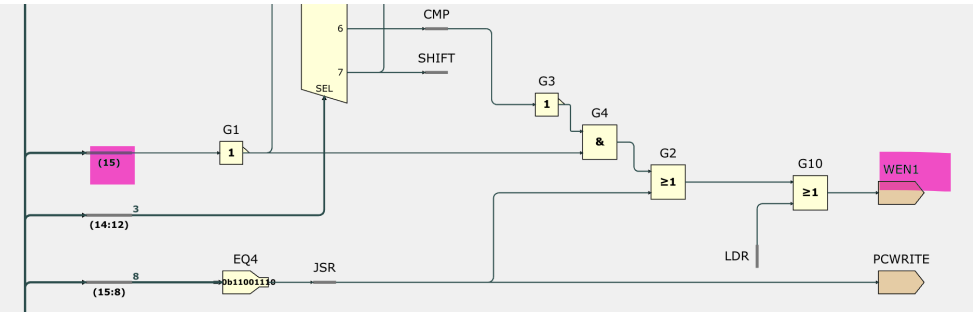
Task 6. Using the material from Lecture 3 and Figure 8, review the behaviour of the CMP instruction in detail and compare it with the other ALU instructions that use ADDSUB.

1. CMP vs SUB. The instruction CMP performs a subtraction internally, but its purpose is to update flags rather than to write a data result.
i. Identify which piece of hardware is responsible for the difference in behaviour between SUB and CMP. In particular, discuss the significance of the signal DPDECODE.WEN1



Difference is in DPDECODE, when CMP chosen, WEN1 will be 0, so that Reg File won't write in anything. When CMP not chosen, WEN1 will be 1 (default value) so that data is written to Reg File.

ii. Inspect the logic in the dpdecode sheet that drives DPDECODE.WEN1. You may assume that, for this question, JSR = 0 and LDR = 0.



(15) is always 0, so it enters G1 and becomes 1. When CMP chosen,
It enters G3 = 0. In G4, 0&1 gives 0, and in G2 and G10, output is still 0.

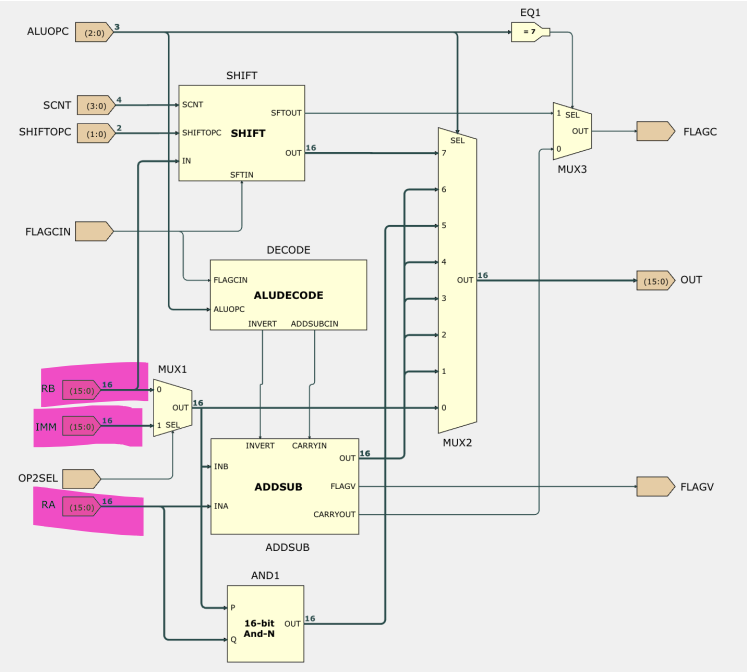
When CMP unchosen, output of G3 = 1, so in G4, 1&1 gives 1. And the final output
Is 1. So Reg File begins to write in numbers.

iii. Write a truth table that shows the value of WEN1 for each of the eight ALU instructions shown in Figure 7. Explain what this table shows about when the ALU result is written back to the register file and when it is not.

ALU instruction	WEN1	Write or not
MOV	1	Yes
ADD	1	Yes
SUB	1	Yes
ADC	1	Yes
SBC	1	Yes
AND	1	Yes
CMP	0	No
SHIFT	1	Yes

2. AND instruction. Look at the multiplexers (MUX2) in the alu sheet and the encoding of the AND instruction in the instruction set.

i. Identify the hardware that implements the AND instruction: which input signals feed the logic block, and through which mux selection does the ALU choose the AND result?



2 inputs, which are RA, and a chosen between immediate and RB enter the AND logic.
And ALU chooses AND at 5 in MUX2.

ii. Explain how this hardware behaviour corresponds to the Boolean AND operation on the two operands.

Component properties

Description

And-N block. 16 And gates used one for each bit of the 16-bit input and output busses.

Number of bits

16

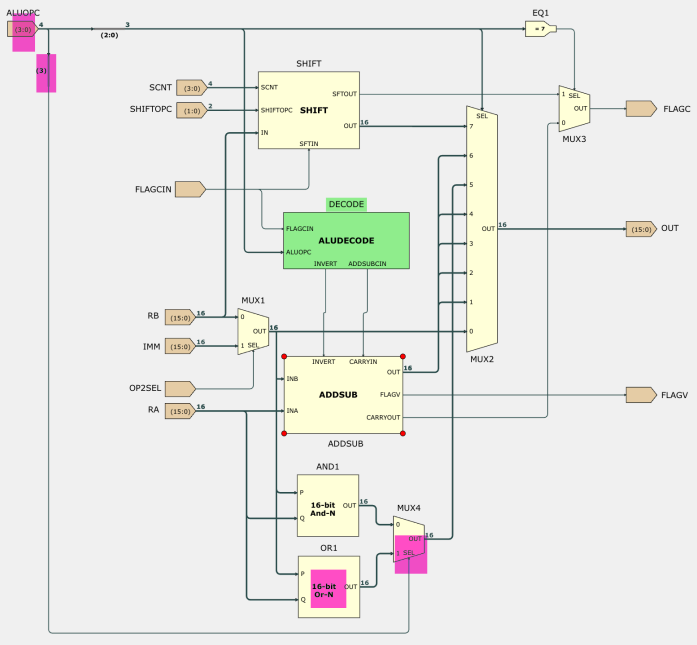
Component Name

AND1

It does Boolean AND operation bit-by-bit, and connects them together and outputs in the end.

1. If you wanted to implement additional logic functions (for example OR, XOR, etc.), which blocks would you need to add or modify? How would you change the existing hardware and the width of ALUOPC to support these new operations?

I need to add: 1. a logic function (like OR, or XOR);
Additional ALUOPC bits as 3-bit can only select at most 8 selections, so an additional bit can select more functions.
An additional MUX to select the new function.



I implemented as following:
1: Add one more bit to ALUOPC, the least 3 bits control MUX2 and other things as before, but the highest bit now does a further selection, to choose whether sends AND or OR function to No.5 in MUX2.
2:Add an OR gate, and connects it to MUX4 with AND gate, then one of their results will be selected and output it selected.

I do 2 tests. In the test, RA=b1111,1111,1111,1111;
RB=b0000,0000,1111,1111;
OP2SEL = 0 so it selects RB;
ALUOPC = 0101 and 1101 separately.

When ALUOPC = 0101, it is AND Gate and OUT is expected to be b0000,0000,1111,1111

hex | dec | bin

Inputs

RA (16 bits)	b1111,1111,1111,1111
ALUOPC (4 bits)	b0101
SHIFTOPC (2 bits)	b00
OP2SEL	0
SCNT (4 bits)	b0000
FLAGCIN	0
RB (16 bits)	b0000,0000,1111,1111
IMM (16 bits)	b0000,0000,0000,0000

Outputs & Viewers

FLAGC	1
FLAGV	0
OUT (16 bits)	b0000,0000,1111,1111

When ALUOPC = 1101, it is OR Gate and OUT is expected to be b1111,1111,1111,1111

hex | dec | bin

Inputs

RA (16 bits)	b1111,1111,1111,1111
ALUOPC (4 bits)	b1101
SHIFTOPC (2 bits)	b00
OP2SEL	0
SCNT (4 bits)	b0000
FLAGCIN	0
RB (16 bits)	b0000,0000,1111,1111
IMM (16 bits)	b0000,0000,0000,0000

Outputs & Viewers

Outputs & Viewers

FLAGC	1
FLAGV	0
OUT (16 bits)	b1111,1111,1111,1111

As expected.